Physics Formulae and Units

Quick revision for JAMB UTME

Motion

- $v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$. Speed = distance/time; velocity = displacement/time; acceleration = change in velocity/time
- Projectile: time of flight $T = 2u \sin\theta / g$; range $R = u^2 \sin 2\theta / g$; maximum height $H = u^2 \sin^2\theta / 2g$ ($g \approx 10\ m/s^2$ in most JAMB questions)

Forces and energy

- $F = ma$; weight $W = mg$; momentum $p = mv$; impulse = Ft = change in momentum
- Work $W = Fd$; power $P = W/t = Fv$; kinetic energy $\frac{1}{2}mv^2$; potential energy mgh; efficiency = useful output/input × 100%
- Hooke's law: $F = ke$. Pressure = force/area; liquid pressure = ρgh. Moment = force × perpendicular distance
- Simple pendulum: $T = 2\pi\sqrt{(l/g)}$. Mass on a spring: $T = 2\pi\sqrt{(m/k)}$
- Gravitation: $F = Gm_1m_2/r^2$

Heat

- $Q = mc\Delta\theta$; $Q = mL$ for change of state. Linear expansion: $\Delta L = \alpha L_o \Delta\theta$
- Gas laws: Boyle's $P_1V_1 = P_2V_2$ (constant T); Charles's $V_1/T_1 = V_2/T_2$ (constant P); pressure law $P_1/T_1 = P_2/T_2$. Combined PV/T = constant. $T(K) = \theta(°C) + 273$

Waves, light and sound

- $v = f\lambda$; $f = 1/T$
- Mirror and lens: $1/f = 1/u + 1/v$; magnification $m = v/u$ = image height/object height. Refractive index $n = \sin i / \sin r$ = real depth / apparent depth
- Sound in air ≈ 340 m/s. Closed pipe fundamental: $\lambda = 4L$; open pipe: $\lambda = 2L$

Electricity and magnetism

- $V = IR$; $P = IV = I^2R = V^2/R$; $Q = It$; energy $E = VIt$
- Resistors in series: $R = R_1 + R_2 + \ldots$; in parallel: $1/R = 1/R_1 + 1/R_2 + \ldots$
- emf: $E = V + Ir$. Capacitors: $C = Q/V$; energy $\frac{1}{2}CV^2$; parallel add, series $1/C = 1/C_1 + 1/C_2$
- Transformer: $Vs/Vp = Ns/Np$. Force on a conductor: $F = BIl$

Modern physics

- Photon energy $E = hf$; $c = f\lambda$. Half-life: $N = N_o\,(½)^{t/T}$. Mass number A = protons + neutrons; atomic number Z = protons

Common units

Quantity	Unit	Quantity	Unit
Force	newton (N)	Pressure	pascal (Pa)
Energy, work	joule (J)	Power	watt (W)
Charge	coulomb (C)	Resistance	ohm (Ω)
Frequency	hertz (Hz)	Capacitance	farad (F)